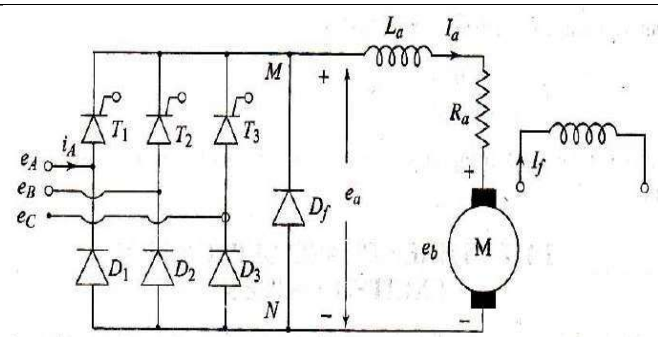
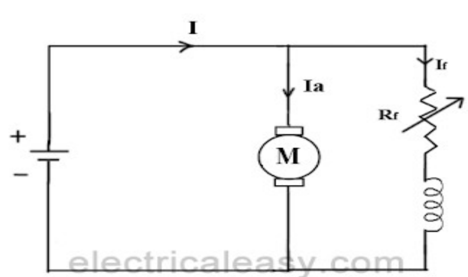
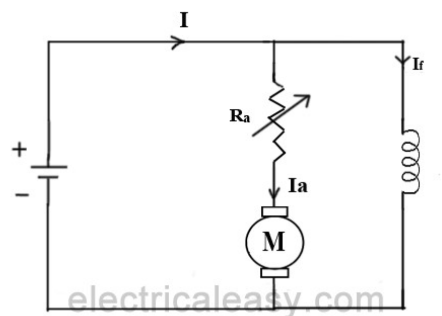
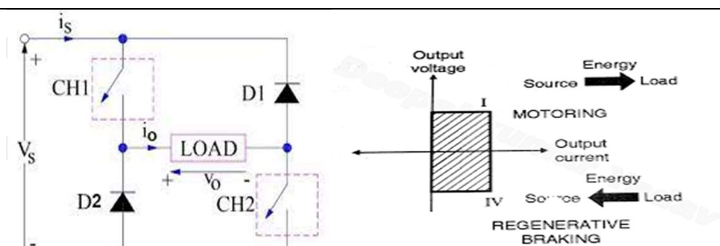
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022			
Course Code: EET413			
Course Name: ELECTRIC DRIVES			
Max. Marks: 100			Duration: 3 Hours

PART A

		Answer all questions, each carries 3 marks.	Marks
1		Advantages ❖ The electrical drives are available in a wide range of torque, speed and power. ❖ They are adaptable to almost any operating conditions such as explosive and radioactive environment, submerged in liquids, vertical mounting and so on. ❖ The electrical drive does not pollute the environment. ❖ It can operate in all four quadrants of speed torque plane. ❖ They can be started instantly and can immediately be fully loaded. Disadvantages ❖ The power failure completely disabled the whole of the system. ❖ The application of the drive is limited because it cannot use in a place where the power supply is not available. ❖ It can cause noise pollution. ❖ The initial cost of the system is high.	(3)
2		Various load torques can be classified into broad categories. Active load torques Passive load torques **1. Active load torques** ❖ Load torques which has the potential to drive the motor under equilibrium conditions are called active load torques. Such load torques usually retain their sign when the drive rotation is changed (reversed).Active torques are closely associated with potential energy.	(3)

		❖ When load is moved upward or a spring is compressed, the stored potential energy increases and active torque developed opposes that actions. That is the torque is directed against the upward movement. ❖ When load is moved downward or a spring is released, the stored potential energy decreases and active torque developed aids that actions. The active torque continue to act in the same direction even after the direction of the drive has been reversed. Examples: 1)Torque due to force of gravity 2) Torque due tension 3)Torque due to compression undergone by an elastic body etc **2)Passive load torques** Load torques which always oppose the motion and change their sign on the reversal of motion are called passive load torques Examples: Torque due to friction, windage, cutting etc.	
3			(3)
4		**Flux Control Method** 	(3)

		The **speed of a dc motor** is inversely proportional to the flux per pole. Thus by decreasing the flux, speed can be increased and vice versa.To control the flux, a rheostat is added in series with the field winding, as shown in the circuit diagram. Adding more resistance in series with the field winding will increase the speed as it decreases the flux. In shunt motors, as field current is relatively very small, I_{sh}^2R loss is small. Therefore, this method is quite efficient. Though speed can be increased above the rated value by reducing flux with this method, it puts a limit to maximum speed as weakening of field flux beyond a limit will adversely affect the commutation. **2. Armature Control Method**  **Speed of a dc motor** is directly proportional to the back emf E_b and $E_b = V - I_aR_a$. That means, when supply voltage V and the armature resistance R_a are kept constant, then the speed is directly proportional to armature current I_a. Thus, if we add resistance in series with the armature, I_a decreases and, hence, the speed also decreases. Greater the resistance in series with the armature, greater the decrease in speed.	
5		 Two quadrant operation of Type D chopper	(3)
6		➢ Regenerative braking control offers second quadrant operation as armature terminal voltage has the same polarity but the direction of I_a is reversed.	(3)

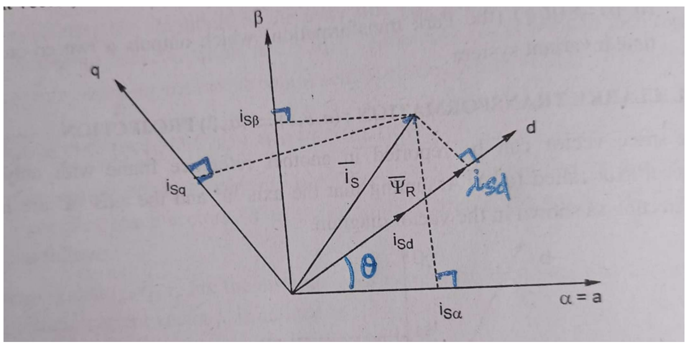
		➢ This configuration is step up Chopper. $V_o = V_s\ (T_{off}\ /\ T)\ = (1- \boldsymbol{\delta})\ V_s$ ➢ Power generated by the motor = $V_o\ I_a$ ➢ Motor emf generated $E_b = K_m\ W_m$ $= V_o + I_a\ R_a$ $=(1- \boldsymbol{\delta})\ V_s\ + I_a\ R_a$ ➢ Motor speed during regenerative braking $W_m = [(1- \boldsymbol{\delta})\ V_s\ + I_a\ R_a\ ]/\ K_m$ 3 marks.	
7		➢ From the expression for the torque developed by an induction motor $\mathbf{T = (3/2\pi n_s)\ [s\ E_2{}^2\ R_2'/\ (R_1^2 + R_2'^2\ )\ + (X_1 + sX_2'^2)]}$ **N-m** ➢ It is directly proportional to the square of the applied terminal voltage at a constant value of supply frequency and slip. By varying the applied voltage, a set of torque-speed curves as shown below can be obtained. When the applied voltage changes by n times the resulting torque changes by n^2 times. ➢ As reduction in supply voltage will reduce the motor torque and therefore the speed of the drive of the motor terminal voltage is reduced to kE_2 , where k<1 , then the motor torque is $\mathbf{T = (3/2\pi n_s)\ [s(\ kE_2\ )^2\ R_2'/\ (R_1^2 + R_2'^2\ )\ + (X_1 + sX_2'^2)]}$ **N-m**	(3)
8		**Advantages of static rotor resistance control** 1. Stepless speed control is possible. 2. Rotor resistance remain balanced between the three phase for all operating points. 3. High starting torque is available at low starting current and improved power factor is possible with wide range of speed control. 4. Maximum torque capability remains unaltered even at speed.	(3)
9		The current space vector represents the three phase sinusoidal system. It needs to be transformed into a two time invariant coordinate system. ❖ This transformation can be divided into two steps:	(3)

- (a, b, c) → (α, β) (the Clark transformation), which gives outputs of two coordinate time variant system.
- (α, β) → (d, q) (the Park transformation), which gives outputs of
- two coordinate time invariant system.

Park transformation

- The two-axis orthogonal stationary reference frame quantities are transformed into rotating reference frame quantities.
- Where θ is the rotor flux position. The flux and torque components of the current vector are determined by the equations.

This is the most important transformation in the FOC. In fact, this projection modifies the two phase fixed orthogonal system (α, β) into d, q rotating reference system.



$$i_{Sd} = i_{S\alpha}\cos\theta + i_{S\beta}\cos\theta$$
$$i_{Sq} = -i_{S\alpha}\sin\theta + i_{S\beta}\cos\theta$$

3 marks. (Note: (a,b,c) → (d, q) direct transformation may also be given full credits.)

10

- The **field oriented control** consists of controlling the stator currents represented by a vector.
- This control is based on projections that transform a three phase time and speed dependent system into a two coordinate (d and q frame) time invariant system.
- These transformations and projections lead to a structure similar to that of a DC machine control.
- FOC machines need two constants as input references:
  the torque component (aligned with the **q coordinate**)
  the flux component (aligned with **d coordinate**).
- The three-phase voltages, currents and fluxes of AC-motors can be analyzed in terms of complex space vectors.

(3)

- If we take i_a, i_b, i_c as instantaneous currents in the stator phases, then the stator current vector is defined as follow:

$$\vec{i_s} = i_a + \alpha i_b + \alpha^2 i_c$$

where, (a, b, c) are the axes of three phase system.

- This current space vector represents the three phase sinusoidal system. It needs to be transformed into a two time invariant coordinate system.

3 marks.

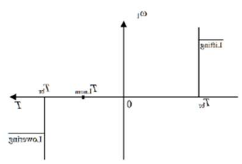
PART B

Answer one full question from each module, each carries 14 marks.

Module I

11 a) (6)

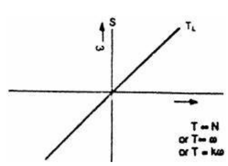
1. **Constant torque load**



- Independent of speed
- T =K
- Ex : Shaping, cutting, grinding
- Requires constant torque irrespective of speed.
- Similarly cranes during the hoisting and conveyors handling constant weight of material per unit time also exhibit this type of Characteristics

2 **Torque is proportional to speed**

- T α w
- T=kw
- Separately excited dc generators connected to a constant resistance load, eddy current brakes have speed torque characteristics given by



3.Torque proportional to square of the speed

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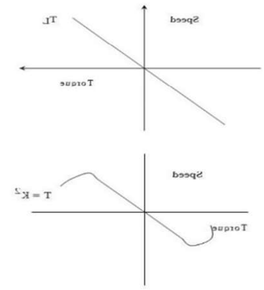
❖ Another type of load met in practice is the one in which load torque is proportional to the square of the speed.



Examples:

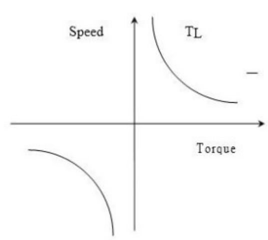
Fans rotary pumps,

Compressors

Ship propellers

4.Torque inversely proportional to square of the speed

❖ Certain types of lathes, boring machines, milling machines, steel mill coiler and electric traction load exhibit hyperbolic speed-torque characteristics



6 marks. Marks to be awarded even if they explain based on different loads like fan, hoist, traction etc.

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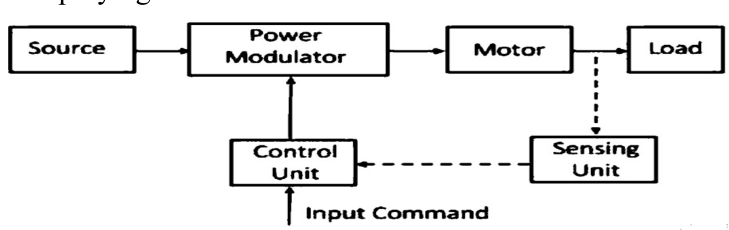
❖ Systems employed for motion control are called as **Drives**.

❖ It may employ any of prime movers such as diesel or petrol engines, gas or steam turbines, steam engines, hydraulic motors and electric motors, for supplying mechanical energy for motion control.

❖ **Drives** employing **electric** motors are called as **Electrical Drives**.



LOAD

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- Load is a machinery designed to accomplish a given task. Examples: fans ,pumps ,robots, washing machines ,trains etc
- Load requirements can be specified in terms of speed and torque demands.
- Motor having speed – torque characteristics and capabilities compatible to the load requirements is chosen.

SOURCES

- Sources may be of 1 phase and 3 phase. 50 Hz AC supply is the most common type of electricity supplied in India, both for domestic and commercial purpose.
- Synchronous motors which are fed 50 Hz supply have maximum speed up to 3000 rpm, and for getting higher speeds higher frequency supply is needed.

Motors of low and medium powers are fed from 400 V supply, and higher ratings like 3.3 kV, 6.6 kV, 11 kV etc are provided also.

POWER MODULATOR

- The power modulator regulates the output power of the source.
- It controls the power from the source to the motor in such a manner that motor transmits the speed torque characteristic required by the load.
- During the transient operations like starting, braking and speed reversing the excessive current drawn from the source. This excessive current drawn from the source may overload it or may cause a voltage drop. Hence the power modulator restricts the source and motor current.
- The power modulator converts the energy according to the requirement of the motor e.g. if the source is DC and an induction motor is used then power modulator convert DC into AC.
- It also selects the mode of operation of the motor, i.e. motoring or braking.
- Power modulators can be classified into three types,

  1) Converters 2) Variable impedance circuits3) Switching circuits.

1) Converters

- Need for converter arises when nature of the available electrical power is different than what is required for the motor.
- Power sources are 1) fixed voltage and fixed frequency AC

2) fixed voltage DC

- Control of dc motors requires variable DC voltage
- Control of AC motors require either fixed frequency variable voltage AC or variable voltage variable frequency AC
- Converters are used to convert currents from one type to another type.
- Depending on the type of function, converters can be divided into 5 types – 1) AC to DC converters 2) AC regulators 3) Choppers or DC-DC converters 4) Inverters 5) Cycloconverters

2) Variable impedance

- Variable resistors are commonly used for the control of low cost DC and AC drives.
- Also needed for the dynamic braking of drives.
- These have two or more steps and can be controlled manually or automatically with the help of contactors.
- Stepless variation of resistors can be obtained using a semiconductor switch in parallel with a fixed resistance.

3) Switching circuits.

- Are required to achieve any one of the following

1. For changing motor connections to change its quadrant of operations.
2. For the changing motor circuit parameter in discrete steps for automatic starting and braking control
3. For operating motors and drives according to predetermined sequence.
4. To provide interlocking to prevent maloperation
5. To disconnect motor when abnormal operating conditions occur.

- Switching operations in motor power circuit are carried out by high power electromagnetic relays known as contactors.

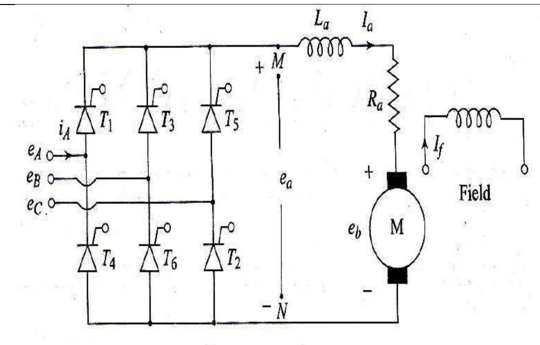
CONTROL UNIT

- The control unit controls the power modulator which operates at small voltage and power levels. The control unit also operates the power modulator as desired.

		❖ It also generates the commands for the protection of power modulator and motor. An input command signal which adjusts the operating point of the drive, from an input to the control unit. **SENSING UNIT** ❖ It senses the certain drive parameter like motor current and speed. ❖ It mainly required either for protection or for closed loop operation. ❖ Performs two functions 1. <u>Speed Sensing</u> ❖ Required for the implementations of closed loop speed control schemes. ❖ Speed is sensed by tachometers ❖ When very high speed accuracies are required as in computer peripherals and paper mills , digital tachometers are used. 2. <u>Current sensing</u> ❖ 2 methods are used. ❖ Current sensor employing hall effect. ❖ Use of non inductive resistance shunt in conjunction with an isolation amplifier which has an arrangement for an amplification and isolation between power and control circuits. Block diagram – 3marks. Explanation – 5 marks.	
		OR	
12	a)	**Dynamics of Motor Load Combination** ❖ J = Polar moment of inertia of motor load system referred to the motor shaft Kg-m^2 W_m = Instantaneous angular velocity rad/sec T = Instantaneous value of developed motor torque N-m T_l = Instantaneous value of load torque referred to motor shaft N-m ❖ Motor torque is the applied torque and load torque is the resisting torque which includes friction and windage torque of the motor. ❖ From the fundamental torque equation :- $T-T_l=d/dt(JW_m)=J(dW_m/dt)+W_m(dJ/dt)$	(6)

		This is applicable to variable inertia drives such as mine winders, reel drives etc. ❖ For drives with constant inertia $dJ/dt=0$ Therefore $T=T_l+J(dW_m/dt)$ ❖ the above equation states that the motor torque is counter balanced by load torque and a dynamic torque $J(d\omega_m/dt)$. ❖ This torque component is termed as dynamic torque as it is only present during the transient operations. ❖ From this equation, we can determine whether the drive is accelerating or decelerating ❖ Drives accelerates or decelerates depending on whether the motor torque is greater or less than load torque. ❖ During acceleration, motor should supply not only the load torque but an additional torque in order to overcome the drive inertia. ❖ In drives with large inertia such as electric trains , motor torque much exceed the load torque by a large amount in order to get enough acceleration. ❖ In drives requiring fast transient response ,motor torque should be maintained at the highest value and motor load system should be designed with lowest possible inertia. ❖ Energy associated with dynamic torque is stored in the form of kinetic energy during acceleration. ❖ During deceleration, dynamic torque has negative sign and maintains drive motion by extracting energy from stored kinetic energy. **6 marks.**	
	b)	In equilibrium steady-state condition, $T_m – T_L = 0$ In Quadrant I $T_m – T_L = 0$ $T_M = 200\text{-}0.2N$ Nm $T_L= 100$ Nm $T_m – T_L = 200\text{-}0.2N -100 = 100 -0.2N = 0$ $N= 500$ rpm	(8)

		In Quadrant II, $T_M = 200\text{-}0.2N$ Nm $T_L = 80$ Nm (unloaded) $T_m - T_L = 0$ $T_m - T_L = 200\text{-}0.2N -(-80) = 0$ $=200 + 80 -0.2N = 0$ N= 1400 rpm In Quadrant III $T_m = -200\text{-}0.2N$ Nm $T_L = -80$ Nm (unloaded) $T_m - T_L = 0$ $= -200\text{-}0.2N -(-80) = 0$ $-200 + 80\ 0.2N = 0$ N=-600 rpm In Quadrant IV $T_m = -200\text{-}0.2N$ Nm $T_L = 100$ Nm (loaded) $T_m - T_L = 0$ $= -200\text{-}0.2N -100 = 0$ $=-300 - 0.2N = 0$ N= -1500 rpm 2 marks each for determining the speed in each quadrant.	
		Module II	
13		Three Phase Full Converter drive connected to a DC separately excited motor: Figure below shows a three phase Full converter drive circuit connected to a DC separately excited DC motor. It is a two quadrant drive without any field reversal. The operation of this circuit is explained with the help of the following important points and with voltage waveforms shown below.	(14)



Important points:

- Thyristors are fired in the sequence of their numbers T1, T2, T3, T4, T5 and T6 with a phase difference of 60 Degrees.
- Thyristors consist of two groups. Positive (Top) group with odd numbered Thyristors T1, T3 &T5 and Negative (Bottom) group with even numbered Thyristors T2, T4 &T6.
- Each thyristor conducts for a duration of 120 degrees and two thyristors conduct at a time one from the Positive group and the other from the Negative group , applying respective line voltage to the motor.
- At any given instant of time, thyristors conduct in pairs and there are six such pairs. They are :

(T6, T1), (T1, T2), (T2, T3), (T3, T4), (T4, T5) and (T5, T6).

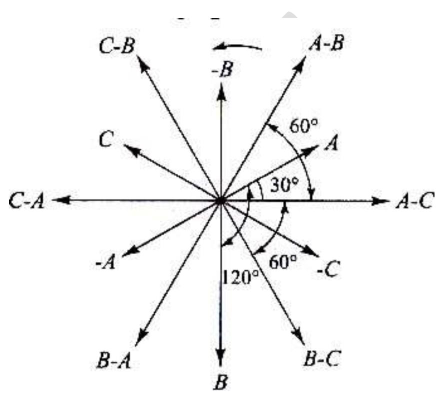
- Each SCR conducts in two consecutive pairs.
- Transfer of current takes place from an outgoing to an incoming thyristor when the respective line voltage is of such a polarity that it not only forward biases the incoming thyristor but it also leads to reverse biasing of the outgoing thyristor when the incoming thyristor turns on. In other words incoming thyristor commutates the outgoing thyristor. i.e T1 commutates T5,T2 commutates T6,T3 commutates T1 and so on.
- The table below gives the details of conducting thyristor pairs, Incoming and outgoing thyristors and the corresponding Line voltages applied across the load.
- For $\alpha > 60°$ the waveforms are different. The voltages go negative due to the inductive load. The previous SCR pair continues to conduct till the next in the sequence is triggered. For e.g. SCRs T6 and T1 continue to conduct up to ($90+ \alpha$) when T2 is triggered. When T2 is triggered it commutates T6 and then the pair (T1,T2) will continue.

- For $\alpha = 90°$ the area under the positive & the negative cycles are equal and the average voltage is zero.
- For $\alpha < 90°$ the average voltage is positive and for $\alpha > 90°$ the average voltage is negative.
- The maximum value of α is 180°
- The output voltage is always a six pulse stream with a ripple frequency of 300 Hz irrespective of the firing angle α.

Table: Firing sequence of SCRs in the 3 φ full converter

S.No.	ωt	Incoming SCR	Conducting pair	Outgoing SCR	Line voltage across the load
1.	$30° + \alpha$	T_1	(T_6, T_1)	T_5	E_{AB}
2.	$90° + \alpha$	T_2	(T_1, T_2)	T_6	E_{AC}
3.	$150° + \alpha$	T_3	(T_2, T_3)	T_1	E_{BC}
4.	$210° + \alpha$	T_4	(T_3, T_4)	T_2	E_{BA}
5.	$270° + \alpha$	T_5	(T_4, T_5)	T_3	E_{CA}
6.	$330° + \alpha$	T_6	(T_5, T_6)	T_4	E_{CB}

- For a better understanding of this topic the vector diagram of the three Phase voltages (w.r.to Neutral) and the six line to line voltages as obtained from a Star connected source are shown in the figure below.



From the above Phasor diagram the Phase and Amplitudes of the three phase voltages and the six line to line voltages can easily be worked out and are given in the table below.

$E_{AN} = E_m \sin(\omega t)$ $\quad E_{BC} = \sqrt{3}\ E_m \sin(\omega t - 90°)$

$E_{BN} = E_m \sin(\omega t - 120°)$ $\quad E_{BA} = \sqrt{3}\ E_m \sin(\omega t - 150°)$

$E_{CN} = E_m \sin(\omega t + 120°)$ $\quad E_{CA} = \sqrt{3}\ E_m \sin(\omega t + 150°)$

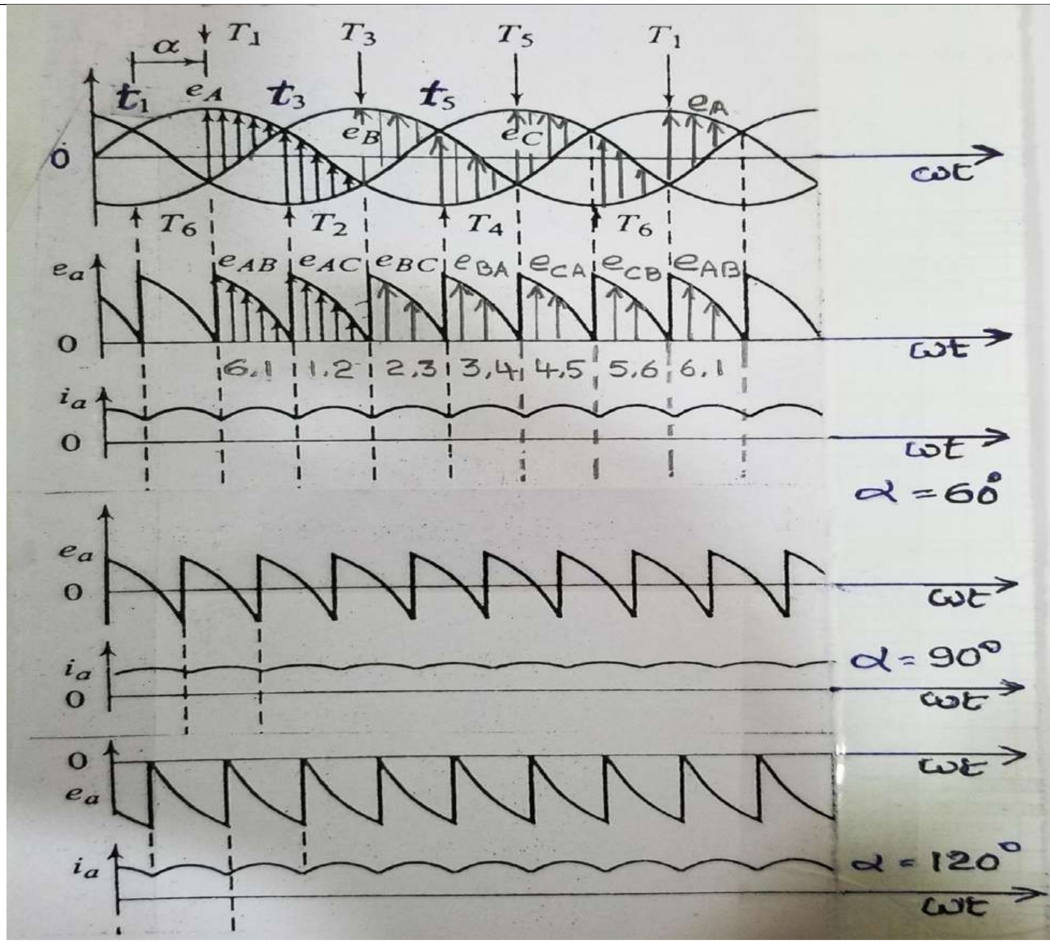
$E_{AB} = \sqrt{3}\ E_m \sin(\omega t + 30°)$ $\quad E_{CB} = \sqrt{3}\ E_m \sin(\omega t + 90°)$

$E_{AC} = \sqrt{3}\ E_m \sin(\omega t - 30°)$

The voltage and current waveforms in this converter for α = 60° are shown in the figure below. The instants of firing the thyristors is marked for α = 60° alone for a clear understanding . The ripple in the output voltage is six pulses per cycle. Since there are six thyristors in the circuit, they are fired at a faster rate (once in 60 °) and the motor current is mostly continuous. Therefore the filtering requirement is less than that in the semi converter system. The operation is explained for the marked firing angle of α = 60°Thyristor T1 turns on at ωt = (30° +. α) . Prior to this SCR T6 was switched ON. Therefore during the interval ωt = (30° + α) to ωt = (30° + α +60°), thyristors T1 and T6 conduct and the Voltage eAB gets applied to the motor terminals. Thyristor T2 gets triggered at ωt = (30° + α +60°) and immediately SCR T6 gets reverse biased and thus gets switched off. The current flow changes from T6 to T2 and so the voltage eAC now gets applied to the motor terminals. This process repeats for every 60° whenever a new thyristor in the sequence gets triggered. The thyristors are numbered in the sequence in which they are triggered.

Applying the same logic the waveform for α = 900 and 1200 is worked out and shown along with that of 600 firing angle. It can be seen that the instantaneous voltages that get applied to the motor become negative for half the period and the average value becomes zero for α = 900. For firing angle of 1200 it can be seen that the amplitude of the negative peaks is larger and the average output voltage is negative.

- As the firing angle α changes from 0 to 90° the output load voltage varies from maximum to zero and the converter is working in Rectifier mode.
- For firing angles of α from 90° to 180° the voltage varies from zero to negative maximum voltage and the converter is working in Inverter mode.

			
		OR	
14	a)	*This is an example where the constant K_a is not given directly and no mention is made about field current. So we can assume field Ø as constant and determine the motor constant $K_aØ$ from the given rated values of voltage (210 V), speed (1200 rpm), armature current (10 A) and the armature resistance (1.5 Ω)* First we have to find out back emf E_b @rated conditions and from that the constant $K_aØ$.	(8)

$E_b = E_a - I_aR_a = 210 - (10\times1.5) = 195$ V

Rated speed in R/s : $\omega = \dfrac{1200 \times 2\pi}{60} = 125.66$ rad/s.

We know that E = K_a Øω , K_a Ø $= \dfrac{E_b}{\omega} = \dfrac{195}{125.66} = 1.55$ V/Rad/sec

Now we can find out the required quantities one by one.

(a) At rated torque, I_a = 10 A.

Back e.m.f. at 800 rpm $= E_{b1} = \dfrac{800}{1200} \times 195 = 130$ V

We know that with continuous conduction required armature voltage $E_a(\alpha)$ from the single phase full converter is given by:

$(\alpha) = \dfrac{2E_m}{\pi}\cos\alpha =$ E I R + E (E = 230 × √2 = 325.27 V)

aa a b m

$$\frac{2\times325.27}{\pi}\cos\alpha = (10\times1.5) + 130$$

From which we get $\alpha = 45.55^0$

(b) For the speed of – 1200 rpm at the rated current E_b = -195 V and the armature loop equation becomes $\dfrac{2\times325.27}{\pi}\cos\alpha = (10 \times 1.5) - 195$

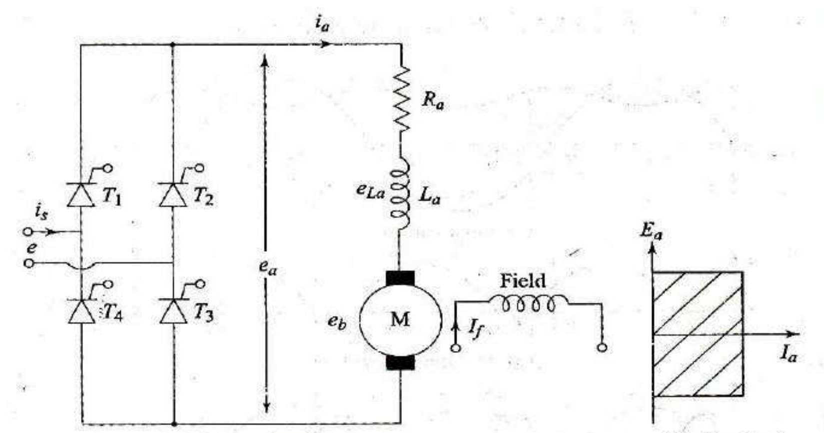
from which $\alpha = 150.37^0$

(c) Substituting the value of armature current I_a at rated torque = 10 A in the armature loop equation and equating it to the converter output at a firing angle of $\alpha = 165^0$, we get . or E = -215.02 V

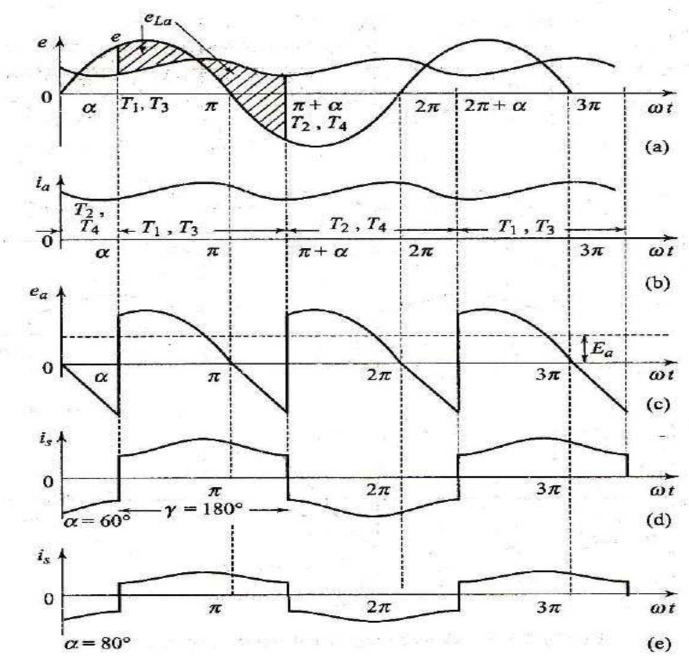
$$\frac{2\times325.27}{\pi}\cos 165^0 = 10\times1.5 + E_b$$

Regenerative braking (second quadrant operation) is obtained either by the field reversal or the armature reversal, for which, K_a Ø = -1.55

Then, $\omega = \dfrac{E_b}{K_a Ø} = \dfrac{-215.02}{-1.55} = 138.72$ rad/s and N = 138.72 x 60/2π = 1325 rpm.

		(3+3+2) marks.	
	b)	A full converter is a two quadrant converter in which the output voltage can be bipolar but the current will be unidirectional since the Thyristors are unidirectional. A full converter feeding a separately excited DC motor is shown in the figure below.  (a) Power circuit (b) Quadrant In this all the four devices are thyristors (T1 to T4) connected in a bridge configuration as shown in the figure.Thyristors T1 and T3 are simultaneously triggered at a firing angle of **α** and thyristors T2 and T4 are triggered at firing angle **$(\pi+\alpha)$**. The voltage and current waveforms under continuous current mode are shown in the figure below. Figure shows the input voltage **e** and the voltage **e_{La}** across the inductance(shaded area).The triggering points of the thyristors are also shown in the figure. The motor is always connected through the thyristors to the input supply. Thyristors T1 and T3 conduct during the interval **$\alpha<\omega t<(\pi+\alpha)$** and connect the supply to the motor. From **$(\pi+\alpha)$** to **α** thyristors T2 and T4 conduct and connect the supply to the motor. At **$(\pi+\alpha)$** when the thyristors T2 and T4 are triggered, immediately the supply voltage which is negative appears across the Thyristors T1 and T3 as reverse bias and switches them off. This is called natural or line commutation. The motor current **i_a** which was flowing from the supply through T1 and T3 is now transferred to T2 and T4. During **α to π** energy flows from the input supply to the motor (both **e & i_s** and **e_a & i_a** are positive signifying positive power flow).However during the	(6)

period **π to (π+ α)** some of the motor energy is fed back to the input system. (**e & i_s** and similarly **e_a & i_a** have opposite polarities signifying reverse power flow)



Circuit diagram -2 marks.

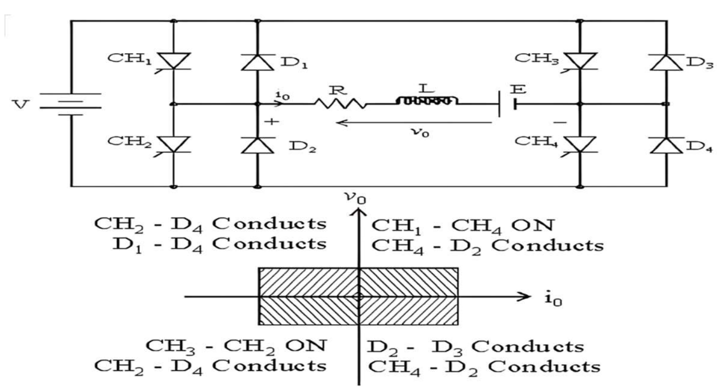
Explanation – 2 marks.

Waveform (output voltage and output current only) - 2 marks.

Module III

15 **Four Quadrant or Class-E Chopper** (14)

- The circuit diagram of a four quadrant or class-E chopper is shown in the figure below.
- It can be considered to be consisting of either two Class-C or Class-D choppers together as shown



FORWARD MOTORING MODE

- When choppers CH1 is operated where as CH4 are kept ON, and CH2 and CH3 are kept off.
- Current flows through the path: E_{DC+}, CH1, load, CH4, E_{DC}-. Since both E_O is positive and positive I_O rises .
- When CH1 is turned off, positive I_O free wheels and decreases as it flows through CH4 and D2 . Hence controlled motor operation in First Quadrant operation.

FORWARD REGENERATIVE BRAKING MODE

- In this mode, motor generated emf is made to exceed the DC source voltage.
- CH1, CH3,CH4 is kept OFF where as CH2 is operated. When CH2 is turned on, negative armature current rises through CH2, D4, E_b , L_a , R_a .
- CH2 is turned off , D1 and D4 are turned ON and motor acts as a generator , returns energy to the DC source.

REVERSE MOTORING MODE

- Opposite to forward motoring mode
- CH1 CH4 are kept OFF and CH2 is kept ON where as CH3 is operated.
- When CH3 and CH2 are on , armature gets connected to source voltage ,so that both armature voltage V_a and armature current I_a are negative.
- As I_a is reversed , motor torque is reversed and consequently motoring mode in third quadrant is obtained.
- When CH3 is turned off, negative armature current freewheels through CH2, D4, E_b , L_a and R_a.

REVERSE REGENERATIVE BRAKING MODE

- This mode is feasible only when $E_b > V_s$
- CH1,CH2, CH3 are kept off, where as CH4 is operated.
- When CH4 is turned ON, Positive Armature current I_a rises through CH4, D2, R_a ,L_a & E_b .

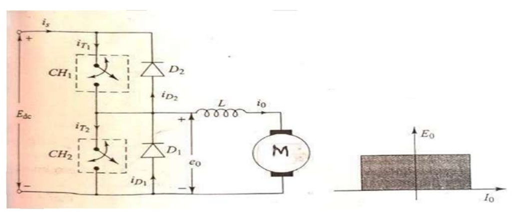
Polarity of E_b must be opposite to that shown. When CH4 is turned OFF , diodes D2, D3 begin to conduct and motor acting as a generator returns energy to the DC Source

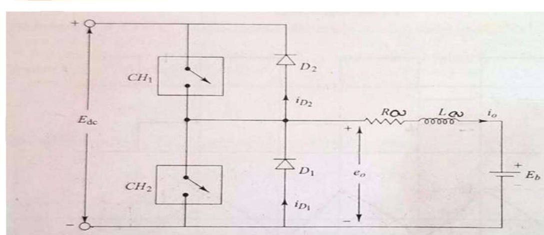
		Circuit diagram and VI characteristics – 4 marks. Quadrant operation explanation – 2.5 marks each.	
		OR	
16	a)	**Solution:** Given Data: V = 220 V, I_a = 24 A, N = 1000 RPM, R_a = 2 Ω V_s = 230 V *(Note: Since the rated voltage of the motor is 220 V which is less than the supply voltage, even for normal operation we have to work with a duty ratio* $\boldsymbol{\delta}$ *of 220 / 230 = 0.956 to ensure that never the applied voltage to the motor exceeds 220 V)* First let us find out E_b for 1000 RPM using the relation V= E_b + I_a R_a i.e. 220 = E_b + 24 x 2 (1 mark) From which we get E_b = 220 – 48 = 172 V for a speed of 1000 RPM Hence for a speed of 500 RPM E_b = (500/1000) x 172 = 86 V (2 marks) Then the required voltage to be applied to the armature is given by: V = (Eb @ 500RPM + Ia @ 1.2 times rated torque x 2) = (86 + 24 x 1.2 x 2) = 143.6 V (2 marks) [Here it is to be noted that the rated current of 24 A is to be multiplied by 1.2 since the motor is now working with a load torque which is 1.2 times the rated torque] Required V for 500 RPM @ 1.2 times rated Torque = 143.6 V *Required duty ratio* $\boldsymbol{\delta}$ *= 143.6/230 = 0.624* (2 marks) Total 7 marks.	(7)
	b)	**Two quadrant (type –A) or class-C chopper** ➢ Class-C chopper can be realised by combining the class-A and class-B choppers in parallel as shown in the figure below. ➢ This combined circuit provides both forward motoring and forward regenerative braking. CH1 along with diode D1 performs forward motoring operation while CH2 along with diode D2 performs the function of forward regenerative braking.	(7)

- Thus for motoring operation CH1 is controlled and for braking operation CH2 is controlled. Shifting of control from CH1 to CH2 shifts operation from motoring to braking and vice versa

MOTORING MODE

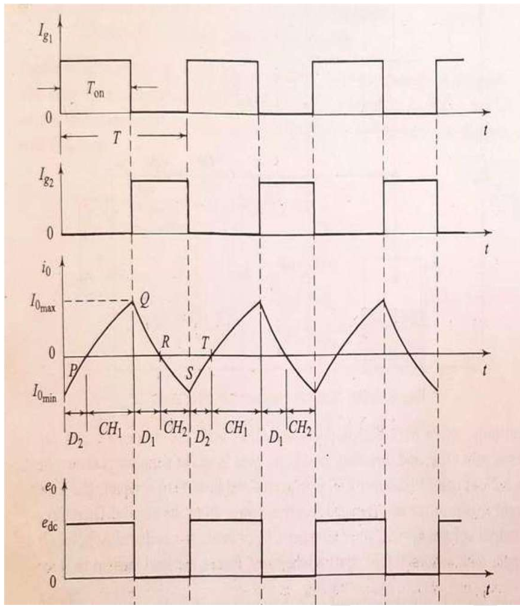
- When CH1 is ON supply voltage V_s gets connected to armature terminals and armature current Ia rises.
- When CH1 is OFF, Ia free wheels through D1 and Ia decays.
- With CH1 and D1 motor control in first quadrant is obtained.
- **Regenerative mode**
- When CH2 is turned on , the motor act as a generator and armature current I_a rises and energy is stored in L_a.
- When CH2 is turned off,D2 gets turned on and the direction of I_a is reversed.
- Energy stored in L_a is returned to DC source and second quadrant operation is obtained.





- Initially when both choppers are OFF, both diodes also are not conducting and hence the load is isolated from the source. As shown in the waveforms above, say initially at point P chopperCH1 is triggered and it starts conducting. The load current is positive and the load receives power from the source. So the output voltage $e_o = E_{DC}$ whenever chopper CH1 conducts.
- At point Q chopper CH1 is turned off, polarity of voltage across inductance L_a changes (becomes negative) and the energy in the inductance forces load current to flow through the diode D1 (in the same direction through the motor i.e.

<table>
<tr><td></td><td></td><td>positive) till the voltage across the inductance L.di/dt becomes equal to the back emf E_b and the load current becomes zero i.e. up to point R.



➢ At this point R, the motor back emf E_b is greater than the voltage across the inductance and since the gate signal for CH2 is present, now E_b forces a current in the opposite direction (negative current) through La and CH2.This continues up to point S i.e. until CH2 is turned off and CH1 is turned on.
➢ Now at point S when CH2 is turned off, polarity of voltage across inductance L_a changes (becomes positive) and the energy in the inductance forces same negative current through the diode D2 into the source until point T when the input current reduces to zero. In this period the current is negative and hence CH1 cannot conduct though it is triggered.
➢ At this point T since gate signal is available to CH1 load current becomes positive, conducts through CH1 and the sequence repeats.
Circuit diagram and VI characteristics – 3 marks.
Explanation with waveform – 4 marks.</td><td></td></tr>
<tr><td></td><td></td><td>Module IV</td><td></td></tr>
<tr><td>17</td><td>a)</td><td>▶ The motor speed can be controlled by varying supply frequency.
▶ Voltage induced in stator is proportional to the product of supply frequency and air-gap flux.</td><td>(7)</td></tr>
</table>

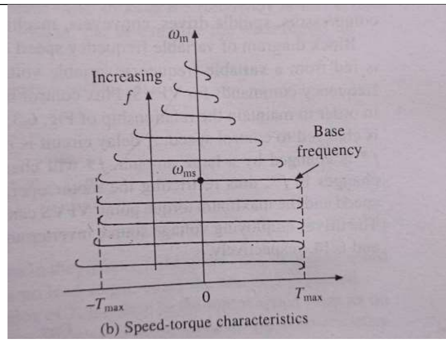
- If stator drop is neglected, terminal voltage can be considered proportional to the product of frequency and flux.
- Any reduction in the supply frequency, without a change in the terminal voltage, causes an increase in the air-gap flux.
- Induction motors are designed to operate at the knee point of the magnetization characteristic to make full use of the magnetic material. Therefore, the increase in flux will saturate the motor.
- This will increase the magnetizing current, distort the line current and voltage, increase the core loss and the stator copper loss, and produce a high-pitch acoustic noise.
- While an increase in flux beyond the rated value is undesirable from the consideration of saturation effects, a decrease in flux is also avoided to retain the torque capability of the motor.
- The variable frequency control below the rated frequency (increase in flux) is generally carried out a rated air-gap flux by varying terminal voltage with frequency so as to maintain (V/f) ratio constant at the rated value.

$$T_{max} = \frac{K(V/f)^2}{\frac{R_s}{f} \pm \left[\left(\frac{R_s}{f}\right)^2 + 4\pi^2 (L_s + L_r')^2\right]^{1/2}}$$

- where K is a constant, and L_s and L_r ‘ are, respectively, the stator and stator referred rotor inductances.
- Positive sign is for motoring operation and negative sign is for braking operation.
- When frequency is not low, $(R_s/f) << 2\pi(L_s + L_r')$

$$T_{max} = \pm \frac{K(V/f)^2}{2\pi(L_s + L_r')}$$

- Above base speed, frequency is changed with V maintained constant, maximum torque decreases with increase in frequency or speed

		 (b) Speed-torque characteristics Equation – 2marks. Explanation – 3 marks. Characteristics – 2 marks.	
	b)	▶ The methods of speed control by voltage control and rotor resistance control have poor efficiency, particularly at low speeds and find limited application. ▶ The slip power is wasted in the rotor resistance, either inherent in the rotor or connected in the rotor circuit. ▶ On the other hand, if the slip power can be returned to the mains, the system can be made efficient and the capacity of the drive can be increased. ▶ The modification of the speed - torque characteristics using a variable rotor resistance has the major disadvantage of poor efficiency, thus making it uneconomical Continuous low speed operation is not possible due to over heating of the motor. ▶ These low speed can very effectively achieved with reasonable efficiency using slip energy or power recovery schemes. ▶ The slip power which is wasted in the external resistance in the rotor circuit is returned to the mains to the schemes. ▶ The conventional methods of slip power recovery employ rotating machines, such as rotary converters, alternators, dc machines, etc. in the rotor circuit to convert the power at conventional schemes, known as Scherbius and Kramer controls. When these methods are employed, the motor may be operated to drive both constant torque and constant power loads. ▶ P_R Rotor input remains constant so long as the motor is supplied for a constant voltage and frequency supply and delivers a load of constant torque and the rotor copper loss P_C is proportional to the slip under this condition.	(7)

- The slip of the motor i.e. its speed, can be altered by extracting power from the rotor and the most common practice was to use a variable external resistance in the rotor circuit. The major drawback of resistance based control is large energy dissipation in the rotor circuit and consequently, low efficiency of the system owing to this disadvantage, speed control by the rotor resistance method is limited in applications to the fields of elevators, cranes etc.
- This above situation is largely improved by utilizing the slip power else where without dissipating it in the external resistance.
- Either the slip power is fed back to the supply source or used to supply an additional motor which is mechanically coupled to the main motor. Thus methodology is termed as slip power recovery scheme it improves the overall efficiency of system to an extremely large extent.
- The speed of the slip ring induction motor can be controlled both in the sub-synchronous and. super -. synchronous regions. This is called cascade connection.
- The slip power is taken from the rotor and feedback to the supply at this condition the motor operates in the sub - synchronous region. Similarly if electric power is pumped to the rotor, the motor operates in the super-synchronous region.
- From the above equation, the motor speed is dependent on the external emf E_{ext}, if ($E_2 - E_{ext}$) is positive. Power flows from the rotor to source of E_{ext} and the motor operates in sub-synchronous region.
- In sub-synchronous operation: Slip power ———-——→ Main source
- If ($E_2 — E_{ext}$) is negative, the power flows from source of E_{ext} to the rotor and the motor operates in the super-synchronous region.
- In super-synchronous operation: Slip power ----------→ Rotor side
- The conventional method of slip power recovery employ rotating machines, such as rotary converters, alternators, dc machines etc. in the rotor circuit to convert the power at slip frequency to power at line frequency.

VARIOUS SLIP POWER RECOVERY SCHEMES

- The slip power recovery system has its classification based on the mode of Operation.

1. Kramer control system

    Conventional Kramer method

    Static Kramer method

2. Scherbius control system

    Conventional Scherbius system

    Static Scherbius system

STATIC KRAMER SYSTEM

- Instead of wasting the slip power in the rotor circuit resistance, it can be converted for 50 Hz ac and pumped back to the line. Here, the slip power can flow only in one direction. This kind of drive is called static Kramer drive.
- The static Kramer system is shown in figure . This system offers speed control only for sub-synchronous speed. i.e., control of speed is possible only when it is less than the synchronous speed.
- In this method, the slip power is taken ‘from the rotor and it is rectified to de voltage by 3-phase diode bridge rectifier as depicted in figure. Inductor L, smoothens the ripples in the rectified voltage V_d . This dc power is converted into ac power by using line-commutated inverter.
- The rectifier and inverter are both line commutated by altering emfs appearing at the slip rings and supply bus bars respectively. Here, the slip power flows from rotor circuit to supply, this method is also called as constant-torque drive.
- The static Kramer drive has been very popular in large power pump and fan type drives, where the range of speed control is limited near, but less than the synchronous speed.
- Static Kramer system of speed control is economical because the rectifier and inverter only have to carry the slip power of the rotor, which is considerably less than the input power to the stator.

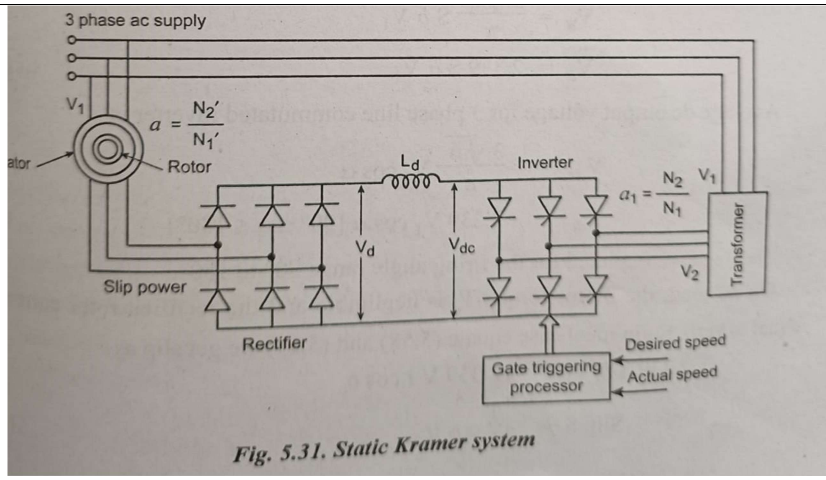


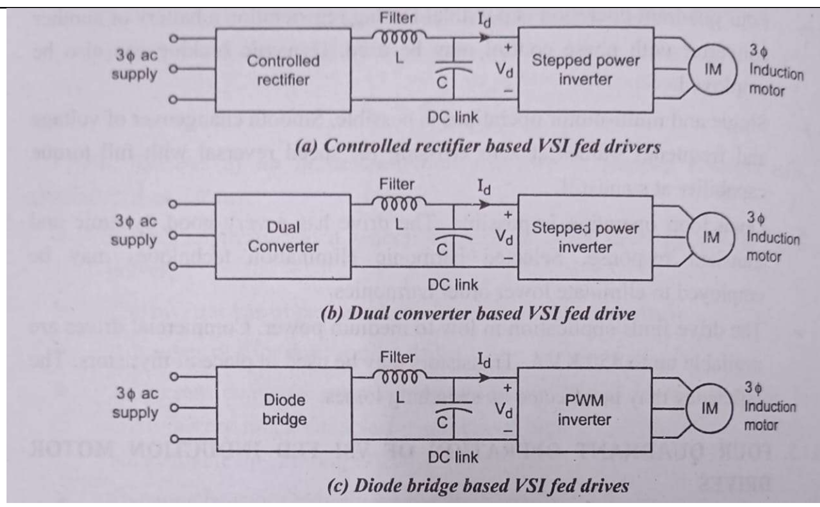
Fig. 5.31. Static Kramer system

Circuit Diagram – 3 marks.

Explanation – 4 marks.

OR

18 a) (9)



(a) Controlled rectifier based VSI fed drivers

(b) Dual converter based VSI fed drive

(c) Diode bridge based VSI fed drives

- Figure (a) :It consists of a 3-phase controlled rectifier, filter and inverter. The 3-phase controlled rectifier converts 3-phase ac supply voltage to variable dc voltage. This voltage is fed to the filter circuit. Here, the inductor L acts as a filter. The output voltage of the filter is fed to the inverter. The inverter produces a variable frequency and variable voltage. The output of the inverter is used to control the input of the ac motor.
- The dual converter based VSI drive is shown in figure (b). It consists of dual converter, filter and inverter. The previous system is not able to regenerate because a reversal of I, would be required. If the regeneration is necessary, the

phase controlled rectifier is replaced by dual converter. The input de voltage is fed to the inverter and is constant due to the capacitor. The inverter output is as variable voltage and variable frequency. The output voltage is fed to the induction motor.

- The diode bridge based VSI consists of bridge rectifier, filter and PWM inverter. The diode bridge rectifier converts ac to fixed dc. This constant dc voltage is fed to the PWM inverter. Here the pulse width modulation techniques are applied in the inverter circuit. The block diagram of this kind is shown in figure (c).

3 marks each with explanation. (Marks to be awarded for chopper based VSI fed drive or direct dc fed to pwm inverter through filter which are other schemes.)

b) (5)

VSI	CSI
VSI is used to fed impressed voltages	CSI is used to fed the impressed currents
Load commutation is not possible	Both forced and load commutations are possible
Open loop and closed loop operations are possible	Only closed loop control is possible
Voltage with variable frequency is fed to Induction motor	The current with variable frequency is fed to Induction motor
The efficiency is medium	The efficiency is high
These drives are suitable for low, medium power applications	These drives are suitable for medium and high power applications
Multimotor operation is possible	Multimotor operation is not possible

Pulse Width Modulation(PWM) technique is used to reduce the problems of harmonic content, torque pulsations, motor heating etc.	The PWM technique is not used in CSI so the distortions in the output are more.
It gives good and satisfactory behavior in transient and dynamic state.	Poor dynamic behaviour

Any 5 points – 5 marks.

Module V

19 a) (14)

- ❖ Two motor phase currents are measured. These measurements feed the Clarke transformation module. The outputs of this projection are designated $i_{s\alpha}$ and $i_{s\beta}$.
- ❖ These two components of the current are the inputs of the Park transformation that gives the current in the d,q rotating reference frame.
- ❖ The i_{Sd} and i_{Sq} components are compared to the references i_{Sdref} (the flux reference) and i_{Sqref}(the torque reference). At this point, this control structure shows an interesting advantage: if can be used fo control either synchronous or induction machines by simply changing the flux reference and obtaining rotor flux position.
- ❖ As in synchronous permanent magnet motors, the rotor flux are fixed (determined by the magnets) there is no need to create one. Hence, when controlling a PMSM, i_{Sdref} should be set to zero.

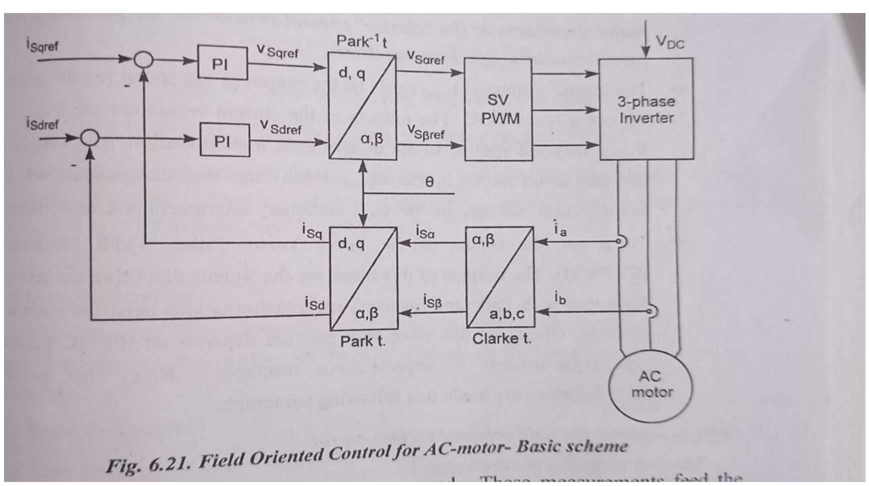
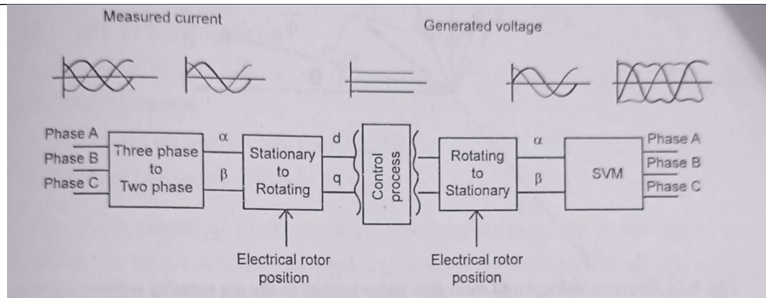


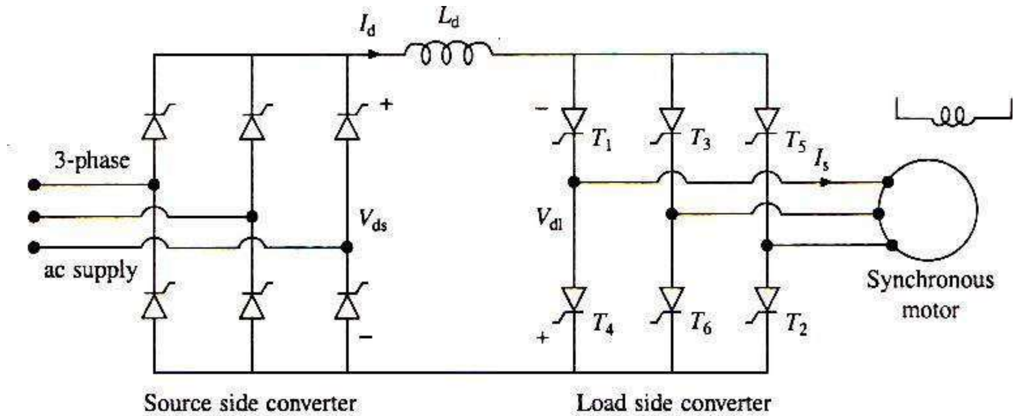
Fig. 6.21. Field Oriented Control for AC-motor- Basic scheme

- As induction motors need a rotor flux creation in order to operate, the flux reference must not be zero. This conveniently solves one of the major drawbacks of the "classic" control structures: the portability from asynchronous to synchronous drives.
- The torque command i_{Sdref} could be the output of the speed regulator when we use a speed FOC.
- The outputs of the current regulators are V_{Sdref} andV_{Sqref} they are applied to the inverse Park transformation. The outputs of this projection are $V_{S\alpha ref}$ and $V_{S\beta ref}$ Which are the components of the stator vector voltage in the α , β stationary orthogonal reference frame.
- . These are the inputs of the Space Vector Pulse Width Modulation (SVPWM). The outputs of this block are the signals that drive the inverter. Note that both Park and inverse Park transformations need the rotor flux position. Obtaining this rotor flux position depends on the AC machine type (synchronous or asynchronous machine). Rotor flux position considerations are made in a following paragraph

Steps to perform for field oriented vector control

1. Measure the motor phase currents
2. Transform them into the two-phase system (α , β) using Clarke transformation
3. Calculate the rotor position angle Transform stator currents into the d,q-coordinate system using Park transformation .
4. The stator current torque (i_{sq}) and flux (i_{sd}) producing components are controlled separately by the controllers.
5. The output stator voltage space-vector is transformed back from the d,q -coordinate system into the two-phase system fixed with the stator by inverse Park transformation
6. Using the space vector modulation, the output three-phase voltage is generated.

		 *Fig. 6.22. Basic Principle of field oriented control* Block Diagram- 6 marks Explanation- 8 marks.	
		OR	
20	a)	Modes of Variable Frequency Control Variable frequency control may employ any of the two modes: (i) true synchronous mode or (ii) self-controlled mode, also known as self-synchronous mode. In true synchronous mode, the stator supply frequency is controlled from an independent oscillator. Frequency from its initial to the desired value is changed gradually so that the difference between synchronous speed and rotor speed is always small. This allows rotor speed to track the changes in synchronous speed. When the desired synchronous speed (or frequency) is reached, the rotor pulls into step, after hunting oscillations. Variable frequency control not only allows the speed control, it can also be used for smooth starting and regenerative braking, as long as it 1s ensured that the changes in frequency are slow enough for rotor to track changes in synchronous speed. A motor with damper winding is used for pull-in to synchronism. In self-control mode, the stator supply frequency is changed so that synchronous speed is the same as rotor speed. This ensures that rotor runs at synchronous speed for all operating points. Consequently, rotor cannot pull out of step and hunting oscillations are eliminated. For such applications, the motor may not require a damper winding. In self-control mode, the stator supply frequency is changed in proportion to the rotor speed so that the rotating field produced by the stator always moves at the same speed as the rotor (or rotor field). Since, the voltage induced in the stator phase has a frequency proportional to rotor speed, self-control can be realized by making the stator supply frequency to track the frequency of induced voltage. Alternatively sensors can be mounted on the stator to track the rotor position. These sensors are called rotor position	(6)

		sensors. The frequency of signals generated by these sensors is proportional to rotor speed. Hence, the stator supply frequency can be made to track the frequency of these signals. 3 marks for each mode.	
	b)	**Load commutated CSI fed synchronous motor:** The circuit diagram of a self controlled synchronous motor drive employing a load commutated thyristor Inverter is shown in the figure below. This drive consists of two parts: Source side converter and load side converter.  The source side converter is a 3 phase 6 pulse line commutated fully controlled converter. When the firing angle range is **0° < α_s < 90°** the converter acts as a line commutated fully controlled rectifier. During this mode the output voltage $\mathbf{V_{ds}}$ and output current $\mathbf{I_{ds}}$ are both positive. When the firing angle range is **90° < α_s < 180°** the converter acts as a line commutated fully controlled inverter. During this mode the output voltage $\mathbf{V_{ds}}$ is negative and output current $\mathbf{I_{ds}}$ is positive. When the synchronous motor operates at a leading power factor, thyristors of the load side converter are commutated by the motor induced voltages just as the thyristors in a line commutated converter are commutated by the supply voltages. This is called Load commutation (here load is synchronous motor).Firing (triggering) angles are referred to the induced voltages just like the triggering angles in a line commutated inverter are referred to the supply voltages.	(8)

When the firing angle range is $\mathbf{0° < \alpha_l < 90°}$ the **load side** converter acts as a line commutated fully controlled rectifier. During this mode the output voltage $\mathbf{V_{dl}}$ and output current $\mathbf{I_d}$ are both positive.

When the firing angle range is $\mathbf{90° < \alpha_l < 180°}$ the **load side** converter acts as a line commutated fully controlled inverter. During this mode the output voltage $\mathbf{V_{dl}}$ is negative and current $\mathbf{I_d}$ is positive.

For $\mathbf{0° < \alpha_s < 90°}$ **&** $\mathbf{90° < \alpha_l < 180°}$ and with $\mathbf{V_{ds} > V_{dl}}$ the source side converter acts like a line commutated Rectifier and load side Converter acts like a line commutated Converter causing power to flow from the source to the motor thus giving motoring operation.

When the firing angles are changed such that $\mathbf{90° < \alpha_s < 180°}$ **and** $\mathbf{0° < \alpha_l < 90°}$ the Load Side Converter acts like a line commutated Rectifier and Source Side Converter acts like a line commutated Inverter causing power to flow from the motor to the source thus giving regenerative braking operation.

The magnitude of Torque depends on $\mathbf{(V_{ds} - V_{dl})}$. The motor speed can be controlled by control of line side converter firing angles.

When working as an Inverter, the firing angle has to be less than 180° to take care of commutation overlap and turn off of thyristors. It is common to define a commutation lead angle for load side converter as

$$\beta_l = 180° - \alpha_l$$

If commutation overlap is ignored, the input AC current of the converter will lag behind the input AC voltage by an angle $\mathbf{\alpha_l}$**.** Since motor input current has an opposite phase to converter input current, the motor current will lead its terminal voltage by an angle $\mathbf{\beta_l}$**.** Therefore the motor operates at a leading power factor.Lower the value of $\mathbf{\beta_l}$**,** higher the motor power factor and lower the Inverter rating.

In a simple control scheme ,the drive is operated at a fixed value of commutation lead angle $\mathbf{\beta_{lc}}$ for the load side converter working as an Inverter and at $\mathbf{\beta_l = 180°}$ **(or** $\mathbf{\alpha_l = 0°}$**)** when working as a rectifier. *When good power factor is required to minimize*

		converter rating, the load side converter when working as an inverter is operated with ***constant margin angle control.*** Circuit Diagram – 4 marks. Explanation – 4 marks.	
****			